

For example, if you intend to change the all the permissions of each file and folder under a specified directory at once, you can use the `sudo chmod` with `-R`:

```
user@host:/home/user$ sudo chmod 777 -R /path/to/
PQRDirectory
```

```
user@host:/home/user$ ls -l
total 3
-rwxrwxrwx 1 user user 0 Jun 14 11:45 Afile
drwxrwxrwx 2 user user 4096 Jun 14 11:45 Xfolder
-rwxrwxrwx 1 user user 0 Jun 14 11:45 Bfile
```

If you want to assign reasonably secure permissions to files and folders/directories, you should give files a permission of `644`, and directories a `755` permission, since `chmod -R` assigns to both using the `sudo` command, the `find` command, and a pipemill to `chmod` as in the following example:

If you only want to change permission of files under a specified directory-

```
user@host:/home/user$ sudo find /path/to/XYZDirectory
-type f -print0 | xargs -0 sudo chmod 644
```

```
user@host:/home/user$ ls -l
total 3
-rw-r--r-- 1 user user 0 Jun 14 11:48 13 A
drwxrwxrwx 2 user user 4096 Jun 14 11:48 folderX
-rw-r--r-- 1 user user 0 Jun 14 11:48 B
```

If you want to only change permission of directories under a specified directory (including that directory):

```
user@host:/home/user$ sudo find /path/to/XYZDirectory
-type d -print0 | xargs -0 sudo chmod 755
```

```
user@host:/home/user$ ls -l
total 3
-rw-r--r-- 1 user user 0 Jun 14 11:52 A
drwxr--r-- 2 user user 4096 Jun 14 11:52 folderX
-rw-r--r-- 1 user user 0 Jun 14 11:52 B
```

Besides giving you the options to change file permissions, Linux also gives you the option of changing the File Owner and Group using the command. For example, if you want to change the `foobar` file's owner to `PQR`, you need to type the following command:

```
user@host:/home/user$ sudo chown PQR foobar
```

If you want to change the `foobar` file's group to `XYZ`, you can use either `chgrp` or `chown` with this syntax:

```
user@host:/home/user$ sudo chgrp XYZ foobar
user@host:/home/user$ sudo chown :XYZ foobar
```

And, if you want to change the `foobar` file's owner to `PQR` and the group to `XYZ` with a single command, you should use the following command: